Narrow scalar clocks, not level repulsion: a fourth spurious-GOE hazard and a re-audit of metallogenic timing statistics

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Abstract

A companion note documented three ways geophysical data fabricate Gaussian-ensemble (GOE) "level repulsion" in the consecutive spacing ratio $\langle r \rangle$ — declustering, seasonal gating, and the Wishart null. Here we add a fourth, and use it to re-audit a published claim of temporal level repulsion in ore-forming systems. A charge-and-release reservoir is a classical relaxation oscillator, i.e. a renewal process; its inter-event intervals are independent, so it has no spectral rigidity and cannot produce genuine level repulsion. Yet a renewal clock with a narrow interval distribution reproduces *any* value of $\langle r \rangle$ — we show it sweeps the whole range from Poisson (0.386) through GOE (0.531) and GUE (0.603) up to 0.88 purely as a function of the interval coefficient of variation (CV). The value of $\langle r \rangle$ therefore carries no evidence of repulsion by itself. Re-analysing the orogenic-gold geochronology of a published four-province study, we exactly reproduce its reported $\langle r \rangle = 0.678$ and show that (i) it measures within-deposit scatter among geochronometers, not province-scale timing; (ii) it is reproduced by a zero-repulsion scalar clock of CV = 0.38; (iii) its apparent rigidity (shuffle deficit -0.031) is an artifact of within-deposit normalisation, which a zero-repulsion null reproduces identically; and (iv) it is unresolvable, since jittering the ages within their stated ± 5 Myr uncertainties moves $\langle r \rangle$ across the entire Poisson-to-GUE range. We restate the program's organising principle accordingly: a single isolated relaxation-oscillator source yields a scalar renewal clock, not GOE level repulsion, while the superposition of many sources yields Poisson; genuine Wigner–Dyson statistics require a chaotic, strongly-mixed spectrum that a low-dimensional dissipative reservoir does not possess.

1 Introduction

The consecutive spacing ratio $\langle r \rangle = \langle \min(s_i, s_{i+1}) / \max(s_i, s_{i+1}) \rangle$ (Oganesyan & Huse, 2007; Ataş et al., 2013) is an attractive, unfolding-free test for order in a sequence, with reference values 0.386 (Poisson), 0.531 (GOE) and 0.603 (GUE). A companion note (Chen, 2026d) showed that three mundane data operations push $\langle r \rangle$ to GOE-like values without any underlying repulsion: minimum-separation declustering, seasonal gating of event times, and the finite-sampling (Wishart) structure of correlation matrices. The unifying remedy was the permutation (shuffle) test together with the coefficient of variation and an appropriate null.

This note adds a fourth hazard that is both more elementary and, in geoscience timing studies, more consequential: a *narrow scalar clock*. Many proposed "single-source" Earth-system rhythms — volcanic recharge, jökulhlaup charge-and-release, ore-fluid focusing — are physically relaxation oscillators: a reservoir fills and discharges at a threshold, then refills. Such a process is a *renewal* process with independent inter-event intervals. It has, by construction, no sequential correlation and no spectral rigidity, and so cannot generate Wigner–Dyson level repulsion. Yet, as we show, a renewal clock with a sufficiently narrow interval distribution reproduces any value of $\langle r \rangle$, including the GOE and GUE values, purely from its marginal. A reported $\langle r \rangle \approx 0.53$ is therefore not evidence of level repulsion.

We make the point concrete by re-auditing a published study that interpreted $\langle r \rangle$-values of 0.57–0.71 across four metallogenic provinces — including a non-magmatic orogenic-gold case — as GOE/GUE "level repulsion" reflecting a charge-and-release memory (Chen, 2026b). The same program's spatial companion (Chen, 2026a) had already cautioned that declustering fabricates GOE; we show the temporal claim falls to the scalar-clock hazard and to the age uncertainties, and we restate the organising principle so that the surviving, defensible content is cleanly separated from the part that does not hold.

2 The fourth hazard: a narrow scalar clock reproduces any $\langle r \rangle$

Mechanism. Model a charge-and-release source as a gamma renewal process with mean-one intervals and coefficient of variation CV (smaller CV = more regular firing). Figure 1a shows $\langle r \rangle$ as a function of CV: it rises smoothly from the Poisson value at CV ≈ 1 through GOE at CV ≈ 0.52 and GUE at CV ≈ 0.45 to 0.88 at CV $= 0.13$. A renewal clock — with no level repulsion whatsoever — thus reproduces the entire range of "RMT" $\langle r \rangle$-values. The value of $\langle r \rangle$ alone is uninformative about repulsion; it is a one-to-one re-encoding of the interval CV.

Why this is not GOE. Genuine Wigner–Dyson statistics arise from the spectra of chaotic, strongly-mixed systems with the appropriate symmetry (Bohigas et al., 1984); integrable or near-integrable dynamics give Poisson statistics (Berry & Tabor, 1977). A classical relaxation oscillator is the integrable extreme — a limit cycle — and its event sequence is a renewal process, not a Hamiltonian spectrum. It has no mechanism for level repulsion. The two genuine fingerprints of GOE that a scalar clock lacks are (i) a small negative shuffle deficit $\langle r \rangle_{\mathrm{obs}} - \langle r \rangle_{\mathrm{shuf}} < 0$, from the sequential anti-correlation (rigidity) of a repulsive spectrum, and (ii) a number variance that grows logarithmically rather than linearly with interval length. Figure 1b shows the shuffle deficit: renewal clocks of every CV give $\langle r \rangle_{\mathrm{obs}} - \langle r \rangle_{\mathrm{shuf}} \approx 0$, while only the genuine GOE spectrum shows a resolved negative value.

The resolution wall. Crucially, that GOE signature is small (≈ -0.03) and requires a long sequence to detect: the standard error of $\langle r \rangle$ is $\approx \sigma_r/\sqrt{N}$ with $\sigma_r \approx 0.27$, so resolving a -0.03 deficit at 2σ needs $N \gtrsim 300$ intervals. Geochronological event sequences have tens. Below that, a scalar clock and a genuine GOE spectrum are observationally indistinguishable in $\langle r \rangle$, and any GOE claim is unsupported regardless of the point value.

3 Re-audit: orogenic-gold timing statistics

We re-analyse the orogenic-gold dataset of Chen (2026b), which reported $\langle r \rangle = 0.678$ ($n = 22$, "best fit GUE", Poisson excluded) and interpreted it as intrinsic charge-release level repulsion in a non-magmatic ore system. The analysis pipeline groups ages by deposit, forms the within-deposit spacings, normalises them by each deposit's mean spacing, and pools the normalised spacings across the ten deposits that have at least three dated ages.

We reproduce $\langle r \rangle = 0.678$ exactly, and find four problems (Fig. 2).

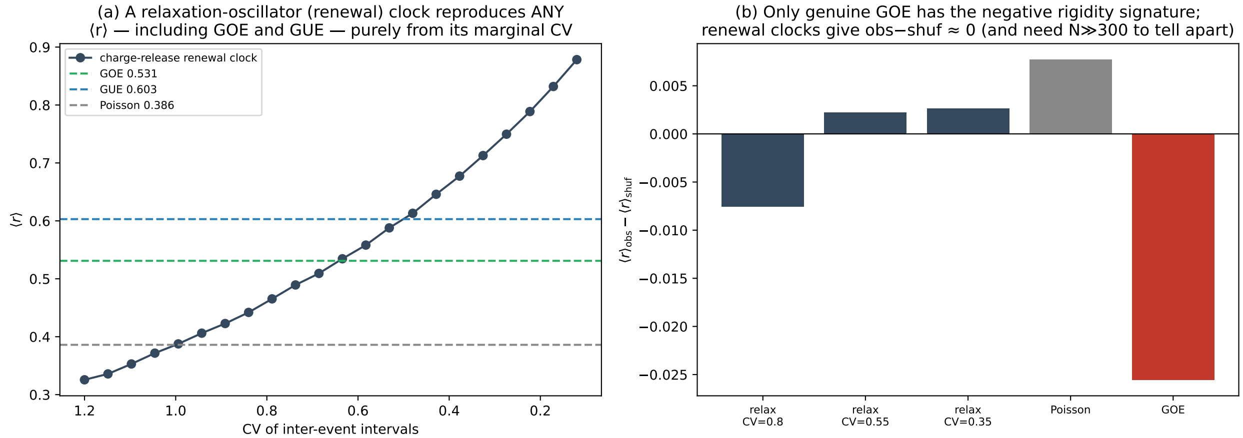

Figure 1: (a) A gamma renewal (charge-release) clock reproduces every $\langle r \rangle$ value — Poisson, GOE, GUE and beyond — purely as a function of its interval coefficient of variation. **(b)** The discriminator: renewal clocks of any CV give a shuffle deficit $\langle r \rangle_{\mathrm{obs}} - \langle r \rangle_{\mathrm{shuf}} \approx 0$; only a genuine GOE spectrum shows the resolved negative (rigidity) signature — and even that needs $N \gtrsim 300$ levels to detect.

(i) It measures the wrong quantity. Each pooled "interval" is the gap between two ages *of the same deposit*, obtained from different geochronometers (U–Pb xenotime, monazite, Re–Os, Ar–Ar). The statistic therefore describes the scatter among dating methods on individual deposits, not the province-scale spacing of distinct mineralising pulses that the charge-release hypothesis is about. There is no province-scale temporal sequence in the construction, so "level repulsion" — a property of a sequence — does not even apply.

(ii) It is a scalar clock. The pooled intervals have CV = 0.31, and a zero-repulsion gamma renewal clock of CV = 0.38 reproduces $\langle r \rangle = 0.68$ (Section 2). The reported value is the marginal CV re-encoded; no repulsion is present or required.

(iii) Its apparent rigidity is a normalisation artifact. The shuffle test gives $\langle r \rangle_{\mathrm{obs}} - \langle r \rangle_{\mathrm{shuf}} = -0.031$, superficially the GOE rigidity signature. But dividing each deposit's intervals by their own mean turns them into *compositional* (closed) data: the normalised intervals of a deposit sum to a fixed value (their count, $\sum_i s_i = \mathrm{const}$), and this closure algebraically forces a negative correlation between adjacent intervals. A pure zero-repulsion renewal clock pushed through the *same* group-normalise-pool construction yields $\langle r \rangle = 0.69$ and a deficit of -0.027 — reproducing both the value and the "rigidity" (Fig. 2b). The -0.031 is an artifact of the closure constraint, not spectral rigidity.

(iv) It is unresolvable. The ages carry ±4–6 Myr uncertainties, comparable to the spacings. Jittering each age within its stated error and re-running the exact pipeline moves $\langle r \rangle$ across $[0.39, 0.67]$ — the full Poisson-to-GUE range — with the reported 0.678 sitting at the upper edge of the noise (Fig. 2a). The published bootstrap interval, which resamples intervals but ignores the age errors, is correspondingly overconfident.

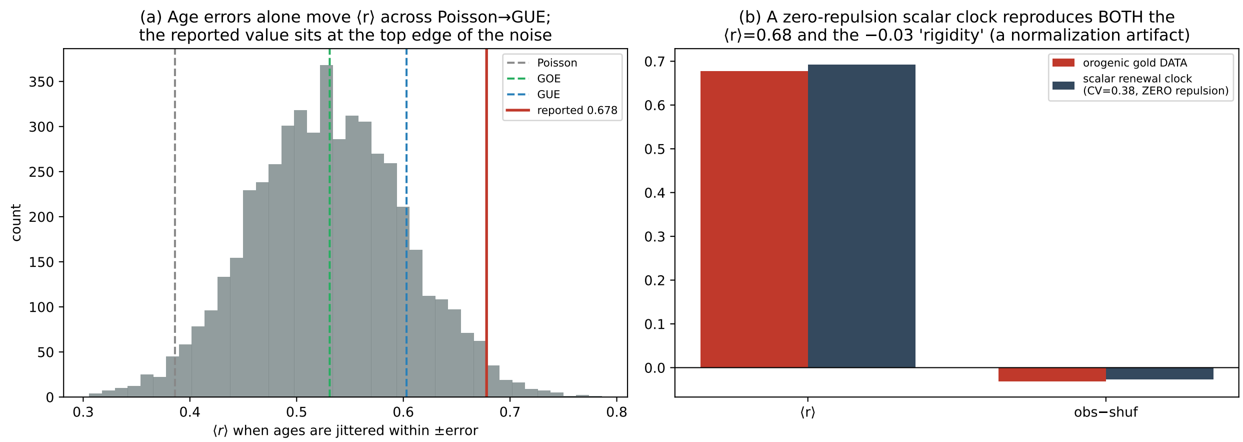

Figure 2: **(a)** Re-running the exact pipeline with the ages jittered within their ±error moves $\langle r \rangle$ across the whole Poisson-to-GUE range; the reported 0.678 is at the top edge. **(b)** A zero-repulsion scalar renewal clock ($\mathrm{CV} = 0.38$) put through the same group-normalise-pool construction reproduces both the $\langle r \rangle = 0.68$ value and the -0.03 shuffle "rigidity": both are construction artifacts, not level repulsion.

4 Restated organising principle

The four points are not a quirk of one dataset; they follow from the mechanism. We therefore restate the program's organising principle (Table 1). A single isolated relaxation-oscillator source — the charge-and-release archetype — produces a *scalar renewal clock*: a narrow interval distribution with high $\langle r \rangle$ set by its CV, shuffle-invariant, and without spectral rigidity. This is *not* GOE level repulsion. Genuine Wigner–Dyson statistics require a chaotic, strongly-mixed spectrum (Bohigas et al., 1984), which a fluid reservoir does not possess. The superposition → Poisson half of the principle is unaffected: pooling independent sources broadens the interval distribution toward exponential and drives $\langle r \rangle \to 0.386$, exactly as observed.

This restatement strengthens rather than weakens the wider program, because it places the ore-timing result in the same family as other correctly-diagnosed scalar clocks — jökulhlaup charge-release and steric nucleosome renewal (Chen, 2026c) — all of which show high, shuffle-invariant $\langle r \rangle$ that is marginal, not sequential.

5 Implications for the audited studies

Metallogenic timing (Chen, 2026b). The central claim — that single ore systems exhibit GOE/GUE level repulsion — is not supported. The high $\langle r \rangle$-values are scalar clocks (and, for orogenic gold, unresolvable given the age errors). What survives is the superposition result: pooling provinces drives $\langle r \rangle$ to the Poisson value, a genuine and correctly-interpreted effect. A corrected version should drop the "level repulsion" interpretation and reframe the single-source signal as a narrow scalar clock, retaining the superposition → Poisson contrast and the data compilation.

Sigma vein spacing (Chen, 2026a). The spatial results of that study — clustered, not stress-shadow-repulsive vein spacing, and the explicit demonstration that declustering fabricates GOE — are sound and unaffected. Only its framing device, a "time–space asymmetry" contrasting temporal repulsion with spatial clustering, rests on the temporal claim audited here. With the temporal side reclassified as a scalar clock, the asymmetry should be restated as "temporal scalar clock plus

Table 1: The organising principle, corrected. The high-$\langle r\rangle$ "single-source" regime is a scalar clock, not level repulsion.

Configuration	Signature	Interpretation
Single relaxation oscillator	high $\langle r\rangle$, CV-set, shuffle-invariant	scalar renewal clock (*not* GOE)
Chaotic / strongly-mixed spectrum	$\langle r\rangle \approx 0.53$, $\langle r\rangle_{\mathrm{obs}} < \langle r\rangle_{\mathrm{shuf}}$, rigid	GOE level repulsion
Superposition of many sources	$\langle r\rangle \to 0.386$	Poisson

spatial clustering", and the spatial and methodological results stand on their own.

6 Conclusion

A narrow scalar (renewal) clock reproduces any value of the spacing ratio, including the GOE and GUE values, with no level repulsion; in short geochronologic sequences it is indistinguishable from a genuine repulsive spectrum, and age uncertainties widen $\langle r\rangle$ across the whole Poisson-to-GUE range. A published claim of temporal GOE/GUE repulsion in ore systems is, on re-audit, a scalar clock measured on within-deposit geochronometer scatter and unresolvable given the age errors. The correct reading of a high, shuffle-invariant $\langle r\rangle$ is a scalar clock, not level repulsion — the necessary fourth control to accompany the no-decluster, de-seasonalise, and density-not-spacing rules of the companion note.

Data and code availability

The orogenic-gold ages re-analysed here are from the dataset of Chen (2026b); a copy and all analysis code are at `https://github.com/Ruqing1963/scalar-clock-not-goe` and archived on Zenodo (DOI assigned on release). All other demonstrations are synthetic.

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